

[This question paper contains 4 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 3005

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Unique Paper Code : 32351402

Name of the Paper : Riemann Integration and
Series of Functions

Name of the Course : B.Sc. (Hons.) Mathematics

Semester : IV

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. All questions are compulsory.
3. Attempt any two parts from each question.
4. Use of calculator is not allowed.

1. (a) Find the upper and lower Darboux integral for $f(x) = 2x + 1$ and show that it is integrable on

[1,2]. Hence show that $\int_1^2 (2x + 1)dx = 4.6.5$

- (b) Prove that a bounded function f on $[a, b]$ is Riemann integrable if and only if it is Darboux

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integrable in which case the values of the integrals agree. (6.5)

(c) Prove that every continuous function f on $[a, b]$ is integrable. (6.5)

2. (a) (i) State and prove intermediate value theorem for integrals.

(ii) Give example of a function f which is not integrable for which $|f|$ is integrable. (6)

(b) If u and v are continuous functions on $[a, b]$ that are differentiable on $[a, b]$ and if u' and v' are integrable on $[a, b]$ then prove

$$\int_a^b u(x)v'(x) + \int_a^b u'(x)v(x) = u(b)v(b) - u(a)v(a) \quad (6)$$

(c) Prove that if f is a piecewise continuous function or a bounded piecewise monotonic function on $[a, b]$ then f is integrable on $[a, b]$. (6)

3. (a) Prove that $\int_0^\infty e^{-t} t^{s-1} dt$ is convergent if and only if $s > 0$. (6)

(b) (i) Prove that $\int_1^\infty \frac{\sin x}{x^2} dx$ converges absolutely

(ii) Prove that $\int_1^{\infty} \frac{\sqrt{x}}{x^3+5} dx$ is convergent. (6)

(c) Test convergence of

(i) $\int_2^{\infty} \frac{2x^2}{x^4-1} dx$ (ii) $\int_0^{\infty} e^{-x^2} x^2 dx$ (6)

4. (a) Let $\langle f_n \rangle$ be a sequence of bounded functions on $A \subseteq \mathbb{R}$. Then this sequence converges uniformly on A to a bounded function f if and only if for each $\epsilon > 0$ there is a number $H(\epsilon)$ in $\mathbb{N}$ such that for all $m, n \geq H(\epsilon)$ then $\|f_m - f_n\|_A \leq \epsilon$. (6.5)

(b) Let $f_n(x) = \frac{nx}{1+n^2x^2}$ for $x \geq 0$, $n \in \mathbb{N}$. Show that the sequence $\langle f_n \rangle$ converges non-uniformly on $[0, \infty)$ and converges uniformly on $[a, \infty)$, $a > 0$. (6.5)

(c) If $a > 0$, show that $\lim_{n \rightarrow \infty} \left(\int_a^{\pi} \frac{\sin nx}{nx} dx \right) = 0$. What happens if $a = 0$. (6.5)

5. (a) State and prove Weierstrass M-test for the uniform Convergence of a series of functions. (6.5)

- (b) Discuss the convergence and uniform convergence of the series of functions

$$\sum (x^n + 1)^{-1}, \quad x \neq 0 \quad (6.5)$$

- (c) If f_n is continuous on $D \subseteq \mathbb{R}$ to $\mathbb{R}$ for each $n \in \mathbb{N}$ and $\sum f_n$ converges to f uniformly on D then show that f is continuous on D . (6.5)

6. (a) (i) Find the radius of convergence of the power series (3)

$$\sum_{n=1}^{\infty} \frac{n^n}{n!}$$

- (ii) Define $\sin x$ as a power series and find its radius of convergence. (3)

- (b) If the power series $\sum_{n=0}^{\infty} a_n x^n$ has radius of convergence R , then the power series

$$\sum_{n=1}^{\infty} n a_n x^{n-1} \quad \text{and} \quad \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1} \quad \text{also have} \quad \text{radius of convergence } R. \quad (6)$$

- (c) Suppose that $f(x) = \sum_{n=0}^{\infty} a_n x^n$ has radius of convergence $R > 0$. Then f is differentiable on $(-R, R)$ and

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \quad \text{for } |x| < R. \quad (6)$$

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Your Roll No.

S. No. of Paper : 6633 HC
Unique paper code : 32351402
Name of the paper : Reimann Integration and Series of Functions
Name of course : B.Sc. (Hons.) Mathematics
Semester : IV
Duration : 3 hours
Maximum marks : 75

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Attempt any two parts from each question.
All questions are compulsory.

Q.1 (a) Prove that a bounded function f on $[a, b]$ is integrable if and only if for each $\varepsilon > 0$, $\exists$ a partition P of $[a, b]$ such that $U(f, P) - L(f, P) < \varepsilon$. (6)

(b) Show that if f is integrable on $[a, b]$, then $|f|$ is integrable on $[a, b]$, and:

$$\left| \int_a^b f \right| \leq \int_a^b |f|.$$

Hence, show that :

$$\left| \int_{-2\pi}^{2\pi} x^2 \sin^8(e^x) dx \right| \leq \frac{16\pi^3}{3}. \quad (6)$$

(c) Suppose f and g are continuous functions on $[a, b]$, and $g(x) \geq 0 \quad \forall x \in [a, b]$. Prove that, $\exists x \in [a, b]$ such that:

$$\int_a^b f(t)g(t) dt = f(x) \int_a^b g(t) dt. \quad (6)$$

Q.2 (a) (i) The Dirichlet function $f: [0, 1] \rightarrow \mathbf{R}$ is defined by :

$$f(x) = \begin{cases} 1, & x \in [0, 1] \cap \mathbf{Q} \\ 0 & x \in [0, 1] \setminus \mathbf{Q} \end{cases}$$

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Is f Riemann Integrable? Justify.

(ii) Show that a decreasing function on $[a, b]$ is integrable.

(b) (i) State Fundamental Theorem of Calculus-I.

(ii) Let f be an integrable function on $[a, b]$. Let P be a partition of $[a, b]$ and P^* be a refinement of P such that $P^* = P \cup \{u\}$. Show that $L(f, P) \leq L(f, P^*)$.

(c) (i) For a bounded function f on $[a, b]$, define the Riemann Sum associated with a partition P . Hence, give Riemann's definition of integrability.

(ii) Calculate $\lim_{h \rightarrow 0} \frac{1}{h} \int_3^{3+h} e^{t^2} dt$

3. (a) Show that:

$$\beta(x, y) = \int_0^1 t^x (1-t)^y dt \text{ converges} \Leftrightarrow x > 0, y > 0.$$

(b) (i) Define Improper Integral of Type-II. When does it converge? When is it said to diverge?

(ii) Determine if $\int_{-\infty}^{\infty} e^x dx$ is a convergent integral. Justify.

(c) (i) Determine if the following integrals are Improper Integrals. If so, what kind? Justify.

$$\int_0^{\pi/2} \frac{\sin x}{\sqrt[3]{x}} dx, \quad \int_0^1 x \ln(x) dx$$

(ii) Prove that $\int_{\pi}^{\infty} \frac{\sin x}{x} dx$ converges absolutely.

4. (a) State and prove Cauchy's criterion for uniform convergence for sequences of functions.

(b) (i) Is the sequence $\{f_n\}$ where :

$$f_n = \frac{1}{n} \sin(nx + n).$$

uniformly convergent on $\mathbf{R}$? Justify. (2)

(ii) Suppose a sequence $\{f_n\}$ converges uniformly to f on a set A , and further suppose that each f_n is bounded on A . Show that the limit function f is bounded on A . (4)

(c) Show that if $a > 0$, then the sequence $\{n^2 x^2 e^{-nx}\}$ converges uniformly on the interval $[a, \infty)$, but that it does not converge uniformly on the interval $[0, \infty)$. (6)

5. (a) If f_n is continuous on $D \subseteq \mathbf{R}$, for each $n \in \mathbf{N}$ and if $\sum f_n$ converges to f uniformly on D , then f is continuous on D . (6)

(b) Show that the series of functions $\sum \frac{x^n}{(1+x^n)}$, $x \geq 0$, converges uniformly on $[0, a]$ for $0 < a < 1$, but is not uniformly convergent on $[0, 1)$. (6)

(c) A sequence $\{f_n\}$ converges uniformly to f on A_0 , if for each $\varepsilon > 0$ there is a natural number $K(\varepsilon)$, depending only on ε , such that if $n \geq K(\varepsilon)$, then:

$$|f_n(x) - f(x)| < \varepsilon \quad \text{for all } x \in A_0.$$

Hence state a necessary and sufficient condition for a sequence $\{f_n\}$ to fail to converge uniformly on A_0 to f .

Apply this condition on sequence $f_n(x) = \frac{x}{n}$, for $x \in \mathbf{R}$ to examine for uniform convergence. (6)

6. (a) (i) Suppose that $f(x) = \sum_{n=0}^{\infty} a_n x^n$ has radius of convergence $R > 0$. Then show that $\int_0^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$ for $|x| < R$.

(ii) State when is a power series differentiable term by term. (1.5)

(b) (i) Apply Abel's Theorem to show :

$$\log_e 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \quad (3.1)$$

(ii) Given $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$. Evaluate:

$$\sum_{n=1}^{\infty} \frac{n}{5^n}, \quad \sum_{n=1}^{\infty} \frac{n^2 (-1)^n}{3^n}. \quad (1)$$

(c) (i) Apply Ratio Test for series to show that radius of convergence R of the power series $\sum a_n x^n$ is given by

$$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|, \text{ whenever the limit exists.} \quad (3.2)$$

(ii) Find radius of convergence for :

$$\sum_{n=2}^{\infty} (\ln(n))^{-1} x^n, \quad \sum_{n=0}^{\infty} \frac{(n!)^3 x^n}{(3n)!}$$

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S. No. of Question Paper : 2815

Unique Paper Code : 32351402

GC-4

Name of the Paper : C-9, Riemann Integration and Series of Functions

Name of the Course : B.Sc. (H) Mathematics

Semester : IV

Duration : 3 Hours

Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

Attempt any two parts from each question.

All questions are compulsory.

1. (a) Prove that a bounded function f on $[a, b]$ is integrable if and only if for each $\varepsilon > 0$, $\exists \delta > 0$ such that :

$$\text{mesh}(P) < \delta \Rightarrow U(f, P) - L(f, P) < \varepsilon,$$

$\forall$ partitions P of $[a, b]$. 6

- (b) Show that every continuous function on $[a, b]$ is integrable. Is the converse true ? Justify. 6

- (c) State and prove Intermediate Value Theorem for integrals. Show, with the help of an example, that the condition of continuity cannot be relaxed. 6

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2. (a) (i) Let f be a bounded function on $[a, b]$. Define the Darboux Integral $\int_a^b f$.

(ii) Give an example of a function f which is not integrable, for which $|f|$ is integrable.

(b) (i) State Fundamental Theorem of Calculus-II.

(ii) Let f be a continuous function on $\mathbf{R}$, and define :

$$F(x) = \int_{x-1}^{x+1} f(t) dt \text{ for } x \in \mathbf{R}$$

Show that F is differentiable on $\mathbf{R}$, and compute F' .

(c) (i) Define :

$$f(x) = [x] \text{ on } [0, 3].$$

Show that f is integrable, and evaluate

$$\int_0^3 f(x) dx.$$

(ii) Define :

$$g : [0, 2] \rightarrow [0, 4] \text{ as } g(x) = x^2.$$

Let :

$$P = \{0, \frac{1}{2}, 1, \frac{3}{2}, 2\}.$$

Find $U(g, P)$.

(a) Show that :

$$\int_0^{\infty} e^{-t} t^{x-1} dt$$

converges $\Leftrightarrow x > 0$.

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(b) Determine the convergence or divergence of :

$$\int_0^{\infty} \frac{1}{e^x \sqrt{x}} dx.$$

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(c) In the following, determine the values of r for which the integral converges/diverges. Determine the values of integrals in case of convergence.

(i) $\int_1^{\infty} x^{-r} dx$

(ii) $\int_0^{\infty} x^{-r} dx$

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(a) Let $\{f_n\}$ be a sequence of integrable functions on $[a, b]$ and suppose that $\{f_n\}$ converges uniformly on $[a, b]$ to f . Show that f is integrable.

6

(b) Let

$$f_n = \frac{1}{(1+x)^n}$$

for x in $[0, 1]$. Find the point-wise limit f of the sequence $\{f_n\}$ on $[0, 1]$.

Does $\{f_n\}$ converge uniformly to f on $[0, 1]$? Justify. 6

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- (c) Show that if $0 < b < 1$, then the convergence of the sequence :

$$f_n = \frac{x^n}{(1+x^n)}.$$

for x in $\mathbb{R}$, $x \geq 0$ is uniform on the interval $[0, b]$, but is not uniform on the interval $[0, 1]$. 6

5. (a) State and prove Weierstrass M-Test and hence examine the uniform convergence of the series :

$$\sum_{n=1}^{\infty} \frac{1}{(x^2 + n^2)}, x \text{ in } \mathbb{R}. \quad 6$$

- (b) Show that the series of functions :

$$\sum \frac{1}{(1+x^n)}, x \neq 0,$$

converges uniformly on $[a, \infty)$ for $a > 1$, but is not uniformly convergent on $(1, \infty)$. 6

- (c) Let $f: A \rightarrow \mathbb{R}$, $A \subseteq \mathbb{R}$ be a bounded function on A . Define uniform norm $\|f\|$ on A . State criterion for uniform convergence of a sequence of bounded functions using uniform norm and hence examine for uniform convergence the sequence : -

$$f_n = x^n; x \in [0, 1]. \quad 6$$

(a) (i) Given :

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

for $|x| < 1$. Evaluate $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$. 2.5

(ii) Let

$$f(x) = \sum a_n x^n$$

for $|x| < R$. If $f(x) = f(-x)$, $\forall |x| < R$, then show that $a_n = 0$ for all odd values of n . 4

(b) (i) If the power series $\sum_{n=0}^{\infty} a_n x^n$ has radius of convergence R , then show that series :

$$\sum_{n=1}^{\infty} n a_n x^{n-1} \quad \text{and} \quad \sum_{n=0}^{\infty} \frac{a_n x^{n+1}}{n+1}$$

also have the radius of convergence R . 4

(ii) Show that :

$$f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad \text{for } x \in \mathbb{R}$$

is well-defined. Also show that :

$$f(x) = f'(x), \quad \text{for } x \in \mathbb{R}.$$

2.5

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- (c) (i) Define interval of convergence for a power series.
Find the radius of convergence and interval of convergence for the following power series :

$$\sum_{n=1}^{\infty} \frac{x^n}{n}, \quad \sum_{n=1}^{\infty} \frac{x^n}{n^2}.$$

- (ii) State Weierstrass's Approximation Theorem.

Name of the Course : **B.Sc. (Hons.) Mathematics CBCS**
 Semester : **IV**
 Unique Paper Code : **32351402_OC**
 Name of the Paper : **C9 - Riemann Integration and Series of Functions**
 Duration: **3 Hours** Maximum Marks: **75**

Attempt any four questions. All questions carry equal marks. All symbols have usual meaning.

1. Calculate the upper Darboux sum and lower Darboux sum of the function

$$f(x) = \frac{1}{x^2} \quad \text{and} \quad g(x) = xe^x \quad \text{on the interval } [1, 2] \text{ for a partition}$$

$$P = \{1, 1.25, 1.5, 1.75, 2\}.$$

Calculate upper and lower Darboux integral of $g(x) = x^2 + 5$ on $[2, 4]$. Is g integrable ?

Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function. Suppose that there is a partition P of $[a, b]$ such that

$$L(f, P) = U(f, P).$$

Show that f is a constant function.

2. Define $f(x) = x[x]$ on $[0, 4]$. Show that f is integrable, and evaluate $\int_0^4 f(x) dx$. Give an example where the functions f and g are not Riemann integrable, but $f \cdot g$ is integrable. Let f be a continuous function on $\mathbb{R}$ and define

$$F(x) = \int_0^{x^4} f(t) dt \quad \text{for } x \in \mathbb{R}$$

show that F is differentiable on $\mathbb{R}$ and compute F' .

3. Examine the convergence of following improper integrals

$$\int_0^\infty e^{-x} (3x + 2) dx \quad \text{and} \quad \int_0^\infty \frac{dx}{\sqrt{3x^4 + 5x}}$$

Using the properties of Gamma integral find the value of

$$\int_0^\infty x^5 e^{-4x^2} dx$$

4. Let (f_n) be defined by $f_n(x) = 1 - |1 - x^2|^n \forall x \in [-\sqrt{2}, \sqrt{2}]$

Find the pointwise limit of (f_n) on $[-\sqrt{2}, \sqrt{2}]$.

Does the sequence converge uniformly on this interval? Justify your answer. Show that the sequence (f_n) where $f_n(x) = nxe^{-nx^2}$, $x \geq 0$ is not uniformly convergent on $[0, 2]$.

Show that the sequence $\left\{ \frac{\sin(n^2 x^2 + 1)}{n(n+1)} \right\}$ converges uniformly on $\mathbb{R}$.

5. Examine the convergence of the series of functions $\sum f_n$ where $f_n(x) = \frac{1}{1+x^n}$ and show that convergence is non uniform in $(1, \infty)$ and is uniform in $[a, \infty[$, $a > 1$

Show that $\sum_{n=1}^{\infty} \frac{\cos nx}{n^3}$ converges uniformly on R to a continuous function.

Evaluate the integral $\int_0^1 \sum_{n=1}^{\infty} \frac{x}{(n+x^2)^2} dx$

6. Find the radius of convergence and exact interval of convergence for the following power series

$$\sum_{n=0}^{\infty} \frac{4^n}{n5^{n+2}} (x-2)^n \quad \text{and} \quad \sum_{n=0}^{\infty} \left[\frac{3+5(-1)^n}{7} \right]^n x^n$$

Write the power series expansion for the integral of the following function:

$$f(x) = \frac{x^2}{3-x^3}$$

given that $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, x \in]-1,1[.$